BIM
FUNDAMENTALS

Building Information Modeling

Presentation • BIM Fundamentals & Revit vs BIM

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01 What is BIM?

BIM = Building Information Modeling

BIM is a coordination process used throughout the building life cycle to create a 3D model and reliable information.

The core idea is not only modeling — it is sharing information and coordinating the project in a common environment.

02 The 3M Concept

BIM can be understood through three main areas:

• Modeling — creating an intelligent 3D representation. • Management — managing project information and processes. • Collaboration — sharing the model and information among project participants.

If there is no 3D model or the model/information is not shared, the source material identifies it as NOT BIM.

03 BIM vs Traditional Workflow

Traditional workflow: Information is transmitted between departments, which can cause delay and information chaos.

BIM workflow: Departments work on coordinated information and connect the process through a Common Data Environment (CDE).

Result: a more reliable shared project model.

04 BIM Dimensions

The course material presents BIM dimensions from 1D to 10D, with each dimension associated with a specialist purpose.

Key examples shown: • 3D — model / visualization • 4D — time / planning • 5D — cost • 7D — green / sustainability

05 What is Revit?

Revit is a BIM software tool used for design authoring and developing a BIM model.

The source lists Revit alongside other design-authoring tools such as ArchiCAD and Tekla Structure.

Revit helps create model elements such as walls, floors, doors, windows and columns.

06 Revit vs BIM

BIM is the PROCESS / METHODOLOGY.

Revit is a SOFTWARE TOOL.

BIM covers collaboration, information management, coordination, analysis and the project life cycle.

Revit is one tool that can be used to create and work with BIM models.

Therefore: BIM ≠ Revit.

07 BIM Uses

BIM can be applied throughout the facility life cycle to achieve specific objectives.

Examples from the material: • Design authoring • Coordination & clash detection • Phase planning / 4D • Quantities, schedules and estimations • Facility management

The information in the BIM grows in quantity and specificity as the project moves through phases.

08 Coordination & Clash Detection

Coordination = integration between project disciplines/facilities.

Clash detection = identifying conflicts between elements.

The material emphasizes: Coordination is NOT the same as clash detection.

Examples of clash-detection tools listed: Navisworks, Tekla BIMsight, Solibri, BIMcollab and BIM Track.

09 BIM Maturity & LOD

BIM maturity levels in the material:

Level 0 — 2D CAD drafting, little/no collaboration. Level 1 — mixture of 3D/2D CAD with standards and CDE. Level 2 — collaborative working with coordinated information exchange. Level 3 — Building Life Management / integrated web environment.

LOD is presented as Level of Development: 100, 200, 300, 350, 400 and 500.

10 Why BIM Matters

The material highlights benefits such as:

• Better coordination and communication • Fewer errors and changes • Increased productivity • Reduced construction cost and waste • More reliable project information • Better planning and facility management

The course material also cites a potential 7–20% reduction in building life-cycle cost and productivity gains of 20–60%.

11 Final Takeaway

BIM is bigger than a software package.

BIM = people + process + information + technology + collaboration.

Revit is one of the tools used to produce and manage BIM information.

The goal is a coordinated, reliable information environment across the project life cycle.